

Overview

Milestone 3 of the **TraceFinder** project focuses on developing a **deep learning–based scanner identification system** using a **Convolutional Neural Network (CNN)**.

The system is designed to learn **scanner-specific intrinsic artifacts**—such as noise patterns, texture inconsistencies, and sensor characteristics—from scanned document images.

To ensure **model transparency and forensic reliability**, **Grad-CAM (Gradient-weighted Class Activation Mapping)** is applied to visually explain **which regions of the document images influence the CNN’s predictions**.

This milestone demonstrates the **end-to-end pipeline** of:

- CNN model training
- Performance evaluation using a confusion matrix
- Model explainability through visual heatmaps

Objectives

- Train a **CNN classifier** to identify the **source scanner** of a document image
- Achieve **training accuracy $\geq 95\%$** despite limited data
- Analyze classification performance using a **confusion matrix**
- Apply **Grad-CAM** to verify that the model focuses on **scanner artifacts rather than document content**

Dataset Description

- All images are stored directly in the **wiki/** directory
- **Class labels are automatically extracted from filenames**
- Naming convention:

scannerA_doc1.jpg → scannerA

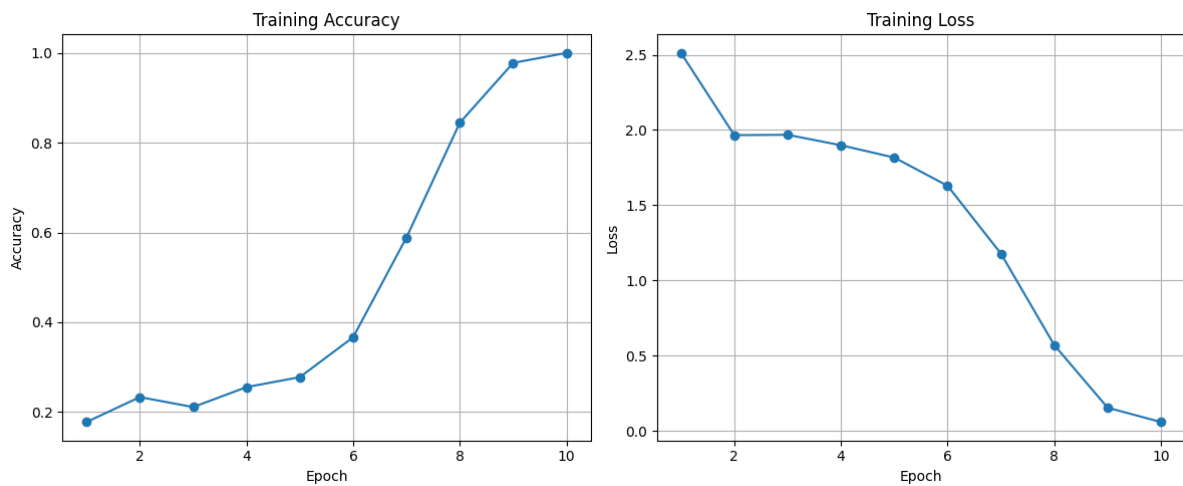
Train the CNN Model

✓ Outputs

- Training accuracy printed in console
- Trained model saved as:

models/cnn_scanner.h5

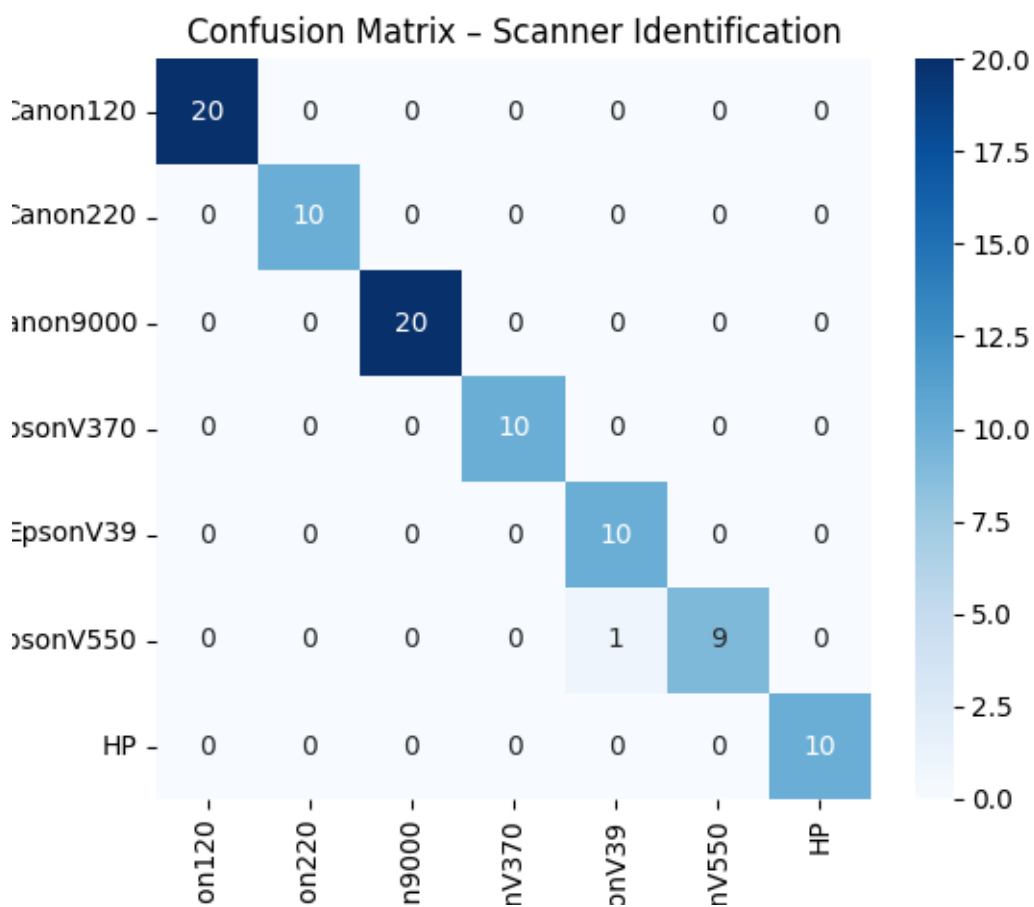
Train accuracy Curve



Generate Confusion Matrix

Outputs

results/confusion_matrix.png



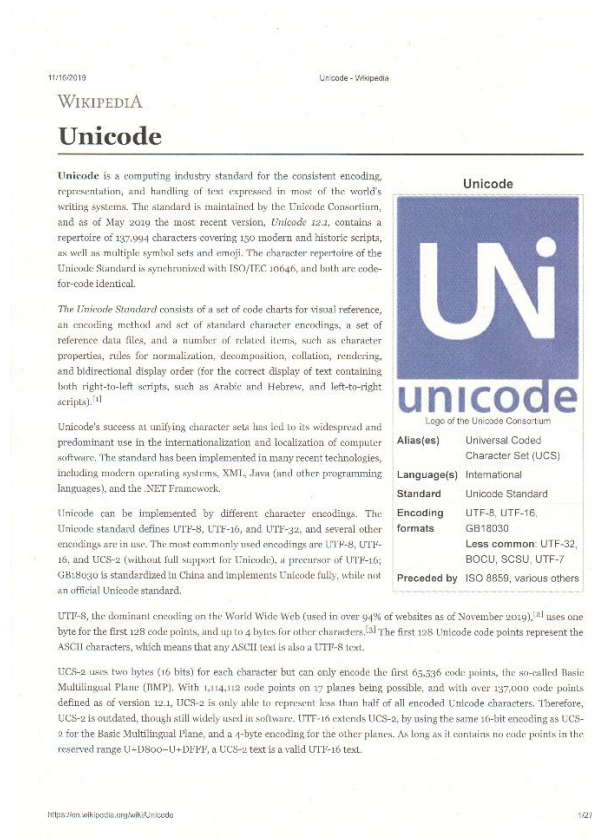
The confusion matrix provides:

- Per-scanner classification performance

- Error distribution across scanner classes

Grad-CAM Explainability

Sample



Outputs

Gradcam_output



Milestone 3 successfully demonstrates:

- Effective **CNN-based scanner identification**

- Strong performance despite limited training data
- Reliable **model explainability** using Grad-CAM

This milestone establishes a **robust foundation for forensic document analysis**, and sets the stage for:

- Larger datasets
- Cross-scanner generalization
- Advanced explainability and attribution methods